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Windows Booting Process

From firmware initialization to the Windows desktop

CPE 22 - System Administration

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LEARNING OBJECTIVES

- Explain the Windows boot sequence
- Compare BIOS and UEFI
- Compare MBR and GPT
- Explain Boot Manager, BCD, the OS loader, and the kernel
- Identify common boot failures

1. Firmware begins the chain

Bootting is a controlled handoff. Firmware tests essential hardware, examines its boot order, and selects a boot device.

Legacy BIOS reads boot code from the Master Boot Record. UEFI can read a filesystem and launch an EFI application directly from the EFI System Partition. UEFI also provides the framework for Secure Boot.

2. MBR and GPT

MBR combines a legacy partition table and first-stage boot code in sector zero. It has practical disk-size and partition-count limits.

GPT stores modern partition metadata with redundancy, supports large disks, and is the normal choice for native UEFI Windows installations.

3. Windows Boot Manager and BCD

Windows Boot Manager reads the Boot Configuration Data store. BCD entries point to Windows installations, recovery tools, devices, and loader options. Boot Manager chooses an entry and launches the associated OS loader.

4. OS Loader and kernel

Winload loads the Windows kernel, Hardware Abstraction Layer, SYSTEM registry hive, and boot-critical drivers. The kernel then initializes memory, scheduling, security structures, and device coordination.

5. Services, sign-in, and desktop

User-mode system processes and services start after the kernel. Windows presents the sign-in experience, loads the selected user profile, and launches the shell that displays the taskbar and desktop.

6. Troubleshooting by stage

No firmware display suggests power, memory, display, or board-level issues. A no-boot-device message points to firmware mode, disk detection, or boot order. BCD errors point to startup configuration. A crash during the logo stage may involve a boot-critical driver or kernel component. A blank desktop after sign-in points later in the chain, often to the profile or shell.

Prefer safe, reversible checks. Record changes, use Windows Recovery Environment where appropriate, and verify current official guidance before destructive repair commands.

Quick reference

Concept	Meaning
BIOS	Legacy firmware that starts boot code from a selected device.
UEFI	Modern firmware that can launch EFI files and support Secure Boot.
MBR	Legacy partition and boot-record format.
GPT	Modern, redundant partition table normally used with UEFI.
Boot Manager	Reads BCD and selects the Windows startup entry.
BCD	Boot Configuration Data store.
OS Loader	Loads the kernel, HAL, registry hive, and critical drivers.
Kernel	Initializes core operating-system resource management.

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